Syllabus

PS50008A Research Methods & Experimental Design | Term 2 Spring 2026

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Module Information

Module Code	PS50008A
Module Title	Research Methods & Experimental Design
Credits	30 (Level 3 / Foundation)
Term	2 (Spring 2026)
Academic Year	2025-26

Module Description

This module introduces foundation students to the principles of psychological research. You will learn how to design experiments, collect and analyse data, and report findings in accordance with professional standards.

A distinctive feature of this module is **DataLab**—in Week 11, you generate the dataset you'll analyse throughout the term. This means you learn statistics using your own data, not abstract examples.

Learning Outcomes

On successful completion of this module, you will be able to:

Conceptual Understanding (C)

- Identify experimental designs (between-participants, within-participants, matched pairs, correlational)
- Distinguish between IVs and DVs
- Formulate appropriate hypotheses (one-tailed, two-tailed)
- Understand null hypothesis significance testing
- Select appropriate statistical tests
- Interpret p-values, confidence intervals, and effect sizes

- Recognise the difference between correlation and causation
- Critically evaluate research claims

Practical Skills (P)

- Calculate descriptive statistics by hand
- Run t-tests and correlations in R
- Produce appropriate data visualisations
- Check statistical assumptions
- Report results in APA format

Research Skills (R)

- Develop research questions
- Operationalise variables
- Design appropriate methodologies
- Collect data systematically
- Write research reports
- Critique methodological approaches
- Consider ecological validity

Teaching Schedule

Session	Day	Time	Location
Lecture	Wednesday	10:00-12:00	RHBD 143
Lab Group A	Thursday	10:00-12:00	WB 304a
Lab Group B	Thursday	10:00-12:00	WB 304b

> **i Week 11**
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> There is no lecture in Week 11. Read the intro lecture reading and attend your lab session on Thursday for DataLab.

Weekly Materials

Each week typically includes:

Material	Description
Lecture Slides	Core content (RevealJS, recorded)
Lecture Reading	Textbook content for the week
Lab Slides	Introduction to lab session
Lab Activity	Interactive WebR exercises

Material	Description
Lab Reading	Supporting practical material
Self-Assessments	Weekly consolidation exercises

Assessment

See the Assessments page for full details.

Assessment	Weight	Deadline
Research Plan	20%	Fri 27 Feb 2026
MCQ Exam	20%	Wed 11 Mar 2026
EDS Assignment	20%	Fri 20 Mar 2026
Research Report	40%	Fri 1 May 2026

Required Reading

Recommended Textbooks

Although I shall be trying to provide detailed readings here, that mean you don't HAVE to lean on textbooks, sometimes a new voice is handy to help grapple with a new topic. Both textbooks below are greaat and are available as e-books via the Goldsmiths Library:

- **Coolican, H. (2017).** ***Research methods and statistics in psychology*** **(6th Ed.).** Hodder and Stoughton. VLeBooks
- **Harris, P. (2008).** ***Designing and reporting experiments in psychology*** **(3rd Ed.).** Open University Press. VLeBooks

Weekly readings are specified on each week's page. See the Resources page for additional materials.

Software

We will talk about this in detail before any use of software is required!

Software	Purpose	Access
R	Statistical analysis	Free download or use the Goldsmiths R Server
WebR	In-browser R exercises	Built into course materials

You do not need to install R to complete this module—all lab activities use WebR, which runs in your browser, and there will be extremely helpful guides and examples.

Group Project

Working in groups is a core part of this module. Later in the term, your group (comprising 4-5 students) will design and run your own study, collect real data, and write it up as a research report. The group project is a substantial part of your assessment, and the quality of your collaboration directly affects your experience and your grade.

Forming Your Groups

In Week 11, you will work in small groups (2-4 students). This matters, so we are having a nice gentle start!

Pay attention to who communicates clearly, who seems enthusiastic, who you can rely on. Consider working with people you feel comfortable working with outside class hours. Your group will work together on labs from Week 16 onwards and on the research project which makes up a large part of this module. It doesn't mean that you can't start planning your group members and group working strategy now though!

> ! Choose Wisely
>
> Group formation is your first opportunity to see how groups naturally assemble and how different people work together. Group work is a VERY psychological phenomenon and the thing we are asked to talk about most often when writing references for jobs or further study.

Project Timeline

Week	Skills & Procedures	Communication
11	Grow your own data	—
12-15	Develop research question and design	Reporting Psychology
—	*Reading Week*	—
16-17	Data collection	Methods & Introduction
18-19	Analysis	Introduction continued
20	Final analysis	Discussion

Assessments from Group Work

The project feeds into two assessments:

- **Research Plan** (due Week 16) — A methods section on a given topic (individual)
- **Research Report** (due Week 22) — Individual write-up of group data (individual)

While the data collection is collaborative, both the Research Plan and Research Report are individual submissions. You are responsible for your own work and your own academic integrity.

ℹ Labs in Project Groups

From Week 16 onwards, labs require you to work in your project groups.

Attendance Policy

Attendance at all lectures and labs is expected and required. This module builds cumulatively—missing sessions will affect your understanding of later material.

- **Attendance** is recorded via SEATS in both lectures and labs
- **Preparation** and independent study are expected each week
- **Preview videos** should be watched before lectures
- **Readings** should be completed as specified

⚠ DataLab

You must attend Week 11 to participate in DataLab. This generates the dataset used throughout the term. If you took the time to contribute the data, the data will play nicely with you! (Guaranteed-ish!)

Academic Integrity

All submitted work must be your own. The group project requires collaboration, but the Research Plan and Research Report are individual submissions.

Using AI tools to generate your work is not permitted. You may use AI to help understand concepts, but submitted work must be your own.

Support

Resource	Details
Office Hours	Thursday 12:00 - 13:00
Email	g.wright@gold.ac.uk
VLE	Discussion board for general questions

Additional Resources

- Resources Page
- Goldsmiths Library
- Academic Skills Centre